

The evolution of speech synthesis models

Large Scale TTS

Table of contents

Introduction

TorToiSe

Scaling TTS

Universal audio generation

Beyond multi-speaker TTS

The majority of the TTS models developed before 2019 were built around one or a few voices, which kept their applications mostly within voice-assistant technologies.

Research focused on improving intelligibility, pronunciation and extending language coverage, with limited success on controllability or voice cloning.

Nowadays, speech synthesis achieves rich diversity and human-level expressiveness, transforming content creation across a wide range of industries.

Improvements

- Prosody and expressiveness

Simpler and better ways to control pace, emphasis, intonation and stress ...

Support paralinguistics like breaths, gasps, filler words (“emh”, “hmm”, “um”) ...

Improvements

- Prosody and expressiveness
- Voice cloning, also known as zero-shot TTS

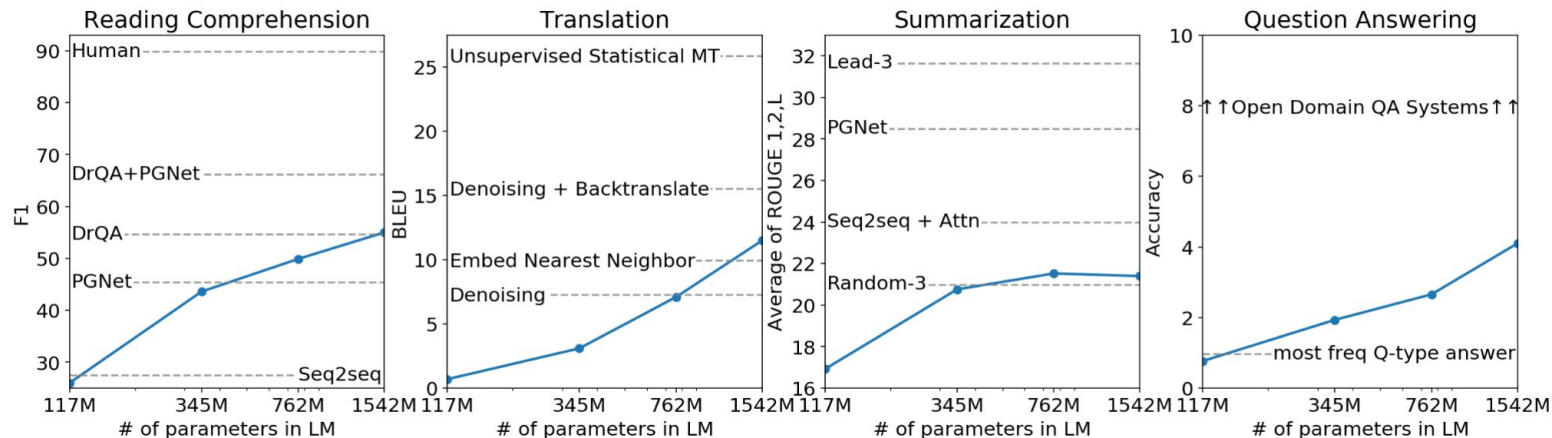
Applied in dubbing, gaming, speech restoration, marketing, education and content creation (audiobooks, podcasts) ...

Improvements

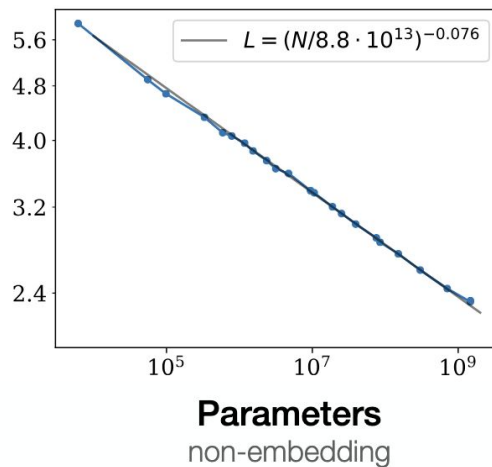
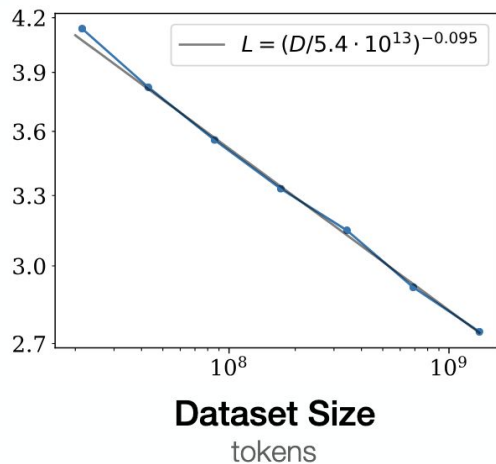
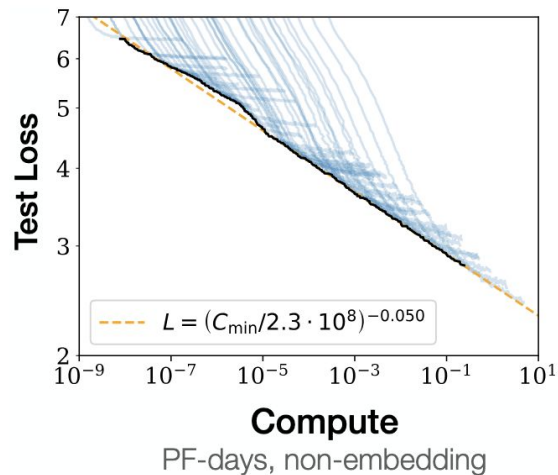
- Prosody and expressiveness
- Voice cloning, also known as zero-shot TTS
- Speaker generation

Meanwhile in GenAI - Large Language Models

In the same year of Transformer TTS (30.7M params) and FastSpeech (30.1M params) generative text models such as GPT-2 scale to more than 1B params.



Meanwhile in GenAI - Large Language Models



Meanwhile in GenAI - Image generation

Could we replicate the same success of DALL-E to zero-shot speech synthesis?

“Could dataset size and model size be the limiting factor of current approaches? In this work, we demonstrate that training a 12-billion parameter autoregressive transformer on 250 million image-text pairs collected from the internet results in a flexible, high fidelity generative model of images controllable through natural language.” [Zero-Shot Text-to-Image Generation](#) OpenAI, Feb 2021



Table of contents

Introduction

TorToiSe

Scaling TTS

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TorToiSe

TTS reaches its DALL-E moment, with TorToiSe, an

- expressive open-source multi-voice TTS system
- supports voice cloning and voice generation

It follows the design of DALL-E, it trades performance for high realistic results, reaching SOTA results.

James Ekter took more than a year to develop TorToiSe as a personal project and he published his work in the technical paper [Better speech synthesis through scaling](#) (2023).

TorToiSe

Design: relies on a high quality speech vocoder and DALL-E's architecture (GPT-2 and speech codecs).

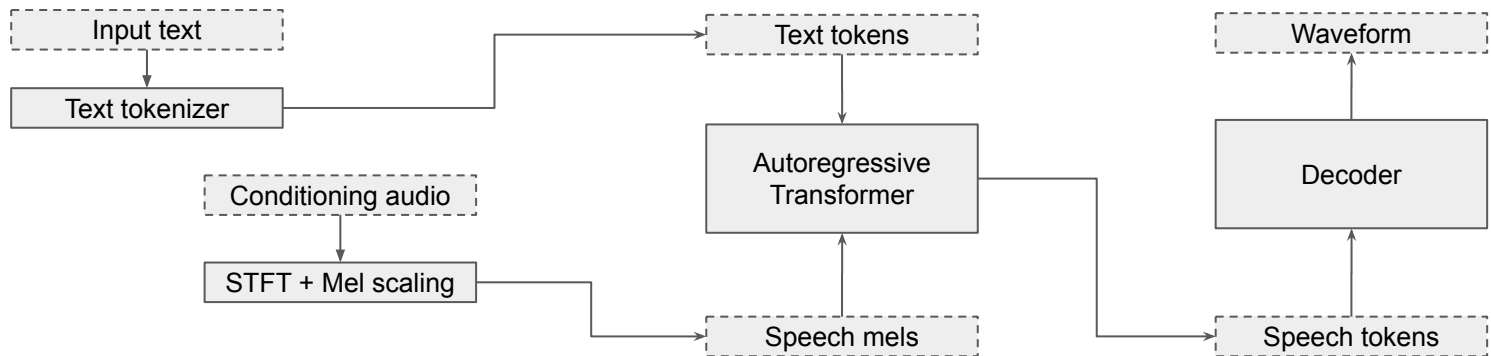
New paradigm: both speech and text are treated as discrete tokens within the transformers paradigm.

Data Scaling: from the usual transcribed public speech datasets (~900 hours) to 49K hours of public data, scraped from the web and transcribed with a custom wav2vec2-large.

TorToiSe

TorToiSe takes as input text and conditioning waveforms to control the speaker identity and style, and it produces speech.

Simplified overview:

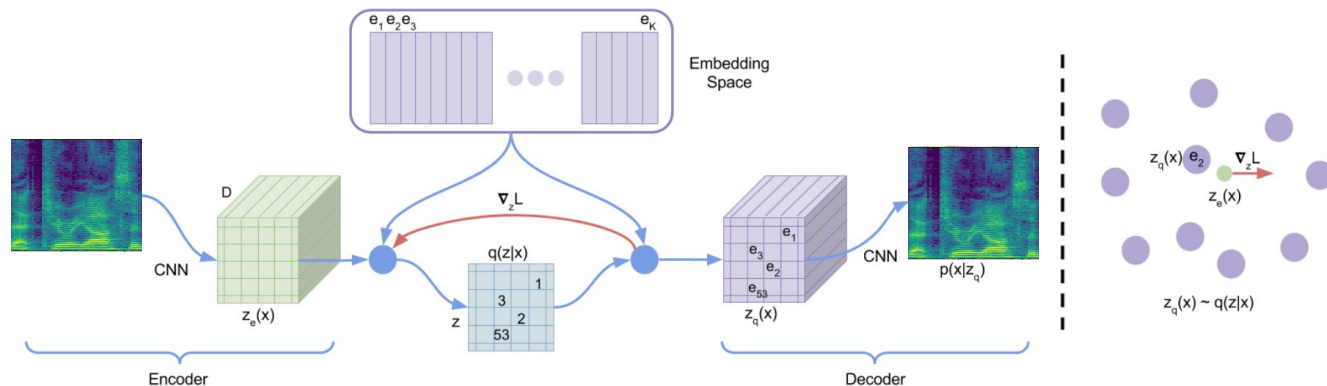


TorToiSe - Main components

The system is made of 5 components which are trained independently:

- **Audio tokenizer**, based on VQ-VAE (van der Oord et al., 2018)

Trained to convert mel-spectrogram into discrete speech codes. A small residual convolutional network that compresses the spectrogram 4 more times and produces a codebook consisting of 8192 tokens.



TorToiSe - Main components

The system is made of 5 components which are trained independently:

- Audio tokenizer
- **Autoregressive decoder**, based on GPT-2 (Radford et al., 2019) as in DALL-E

Conditioned on text tokens and a single vector embedding from 2 reference mel-spectrograms (6 seconds of audio each). Speech (MEL) tokens and the text tokens have their own positional parameters. The maximum sequence length is 402 text tokens + 604 speech tokens. The prompt is assembled as follows:

`<SC><BT><T><T><T>..<T><ET><BM><M><M><M>...<EM>`

SC=Speech conditioning encoding

BT=Begin text token

T=Text tokens

ET=End text token

BM=Begin MEL token

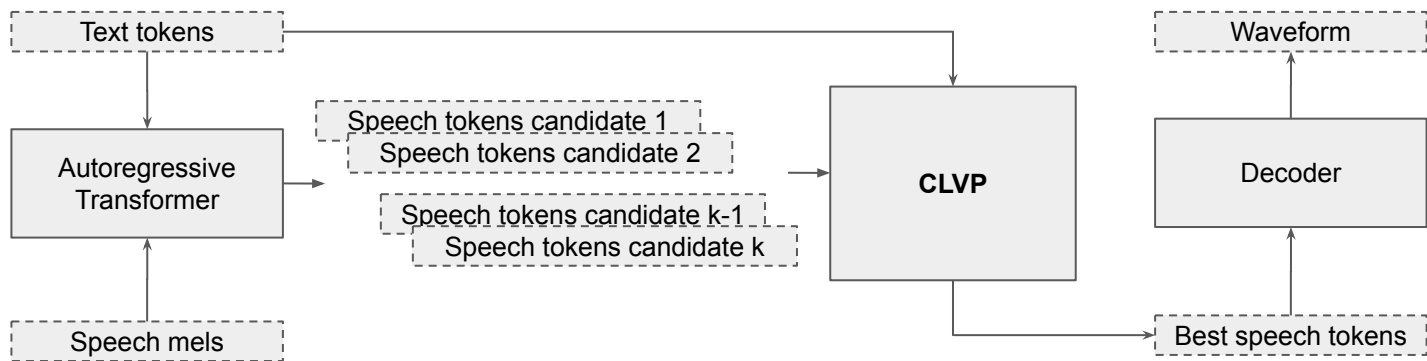
M=MEL tokens

EM=End MEL token

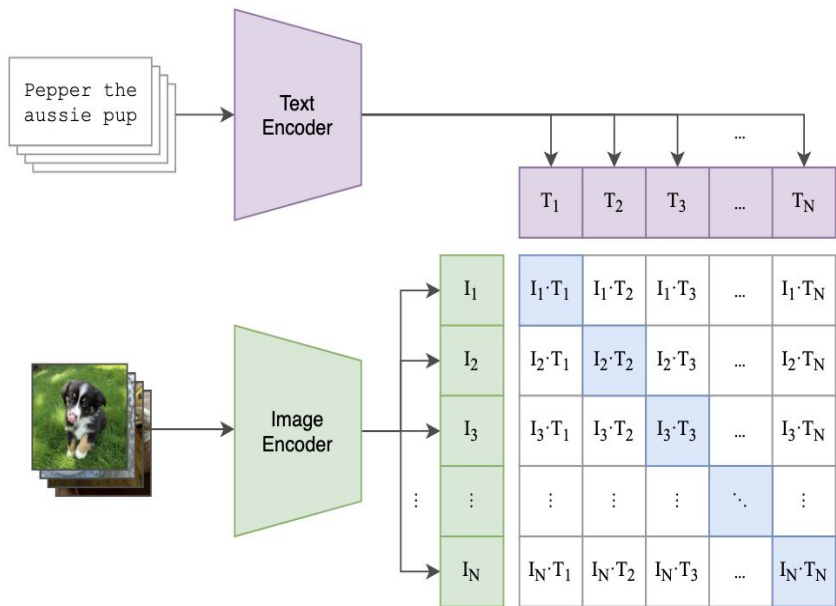
TorToiSe - Main components

The system is made of 5 components which are trained independently:

- Audio tokenizer
- Autoregressive decoder
- **Contrastive Language-Voice Pretrained Transformer (CLVP)**, based on Contrastive Language-Image Pre-training (CLIP) (Radford et al., 2021)



CLIP



Training

Large batches of image-caption pairs and optimises so that correct pairs have high cosine-similarity, while incorrect pairs have low similarity. Trained on 400 million image-text pairs collected from the Internet.

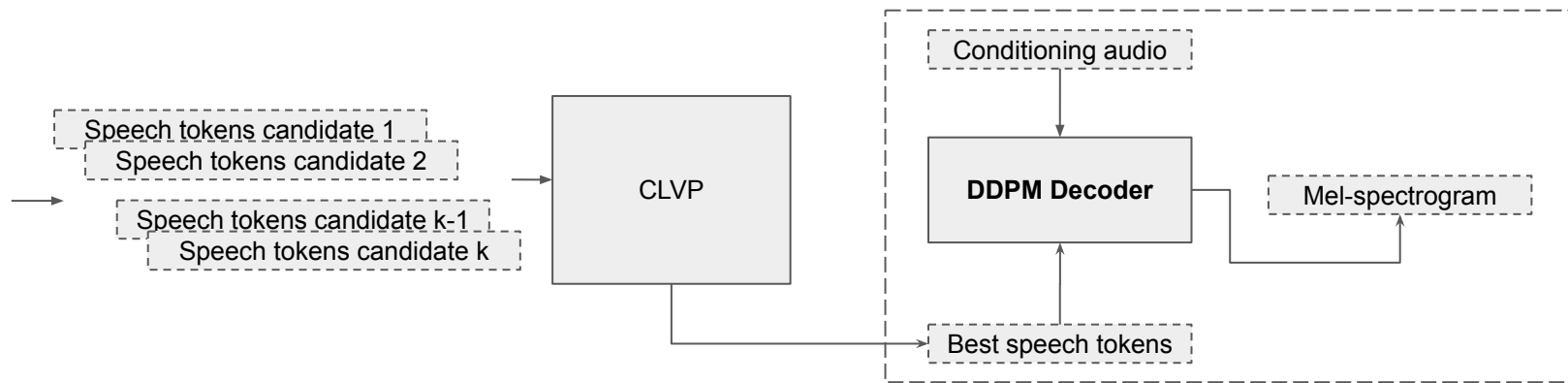
Purpose

Determines how well an image and a text caption go together.

TorToiSe - Main components

The system is made of 5 components which are trained independently:

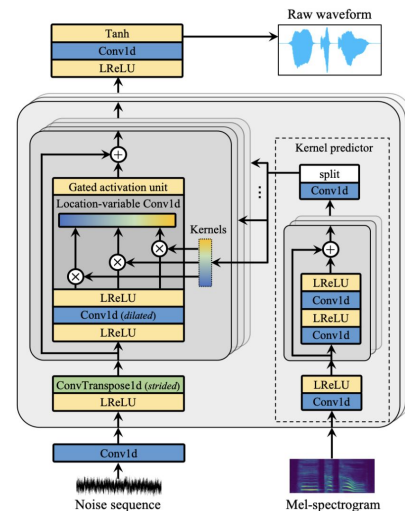
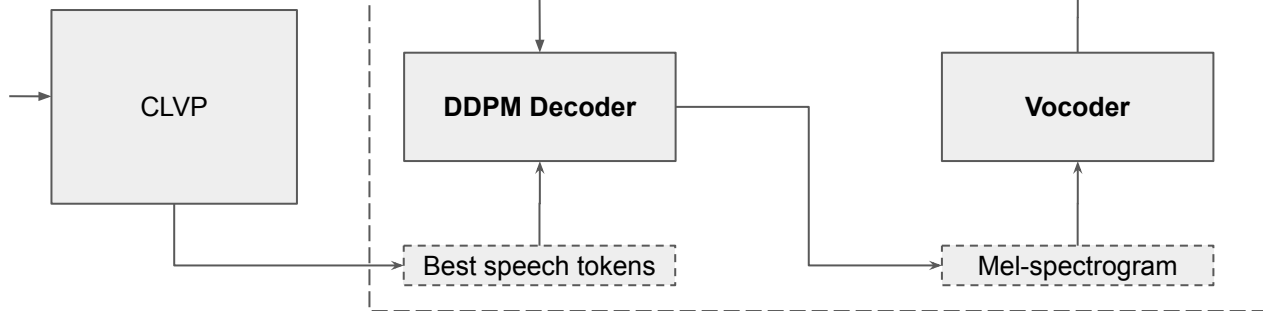
- Audio tokenizer
- Autoregressive decoder
- Contrastive Language-Voice Pretrained Transformer (CLVP), based on Contrastive Language-Image Pre-training (CLIP) (Radford et al., 2021)
- **Denoising Diffusion Probabilistic Models (DDPM) Decoder** (Ho et al., 2020)



TorToiSe - Main components

The system is made of 5 components which are trained independently:

- Audio tokenizer
- Autoregressive decoder
- CLVP
- DDPM Decoder
- **Vocoder, Univnet (Kim, 2021)**



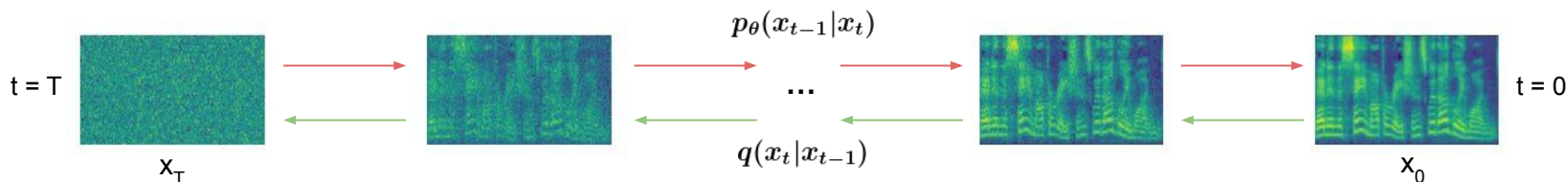
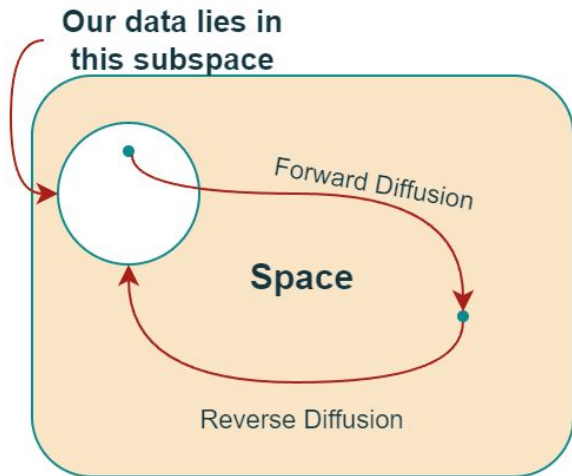
Denoising Diffusion Probabilistic Models (DDPM)

Two Processes:

- **Forward diffusion** - Start with a real image and slowly add Gaussian noise over many steps turning it into pure noise
- **Reverse (denoising) process** - A neural network learns to undo that noise step-by-step, turning noise back into a clean image

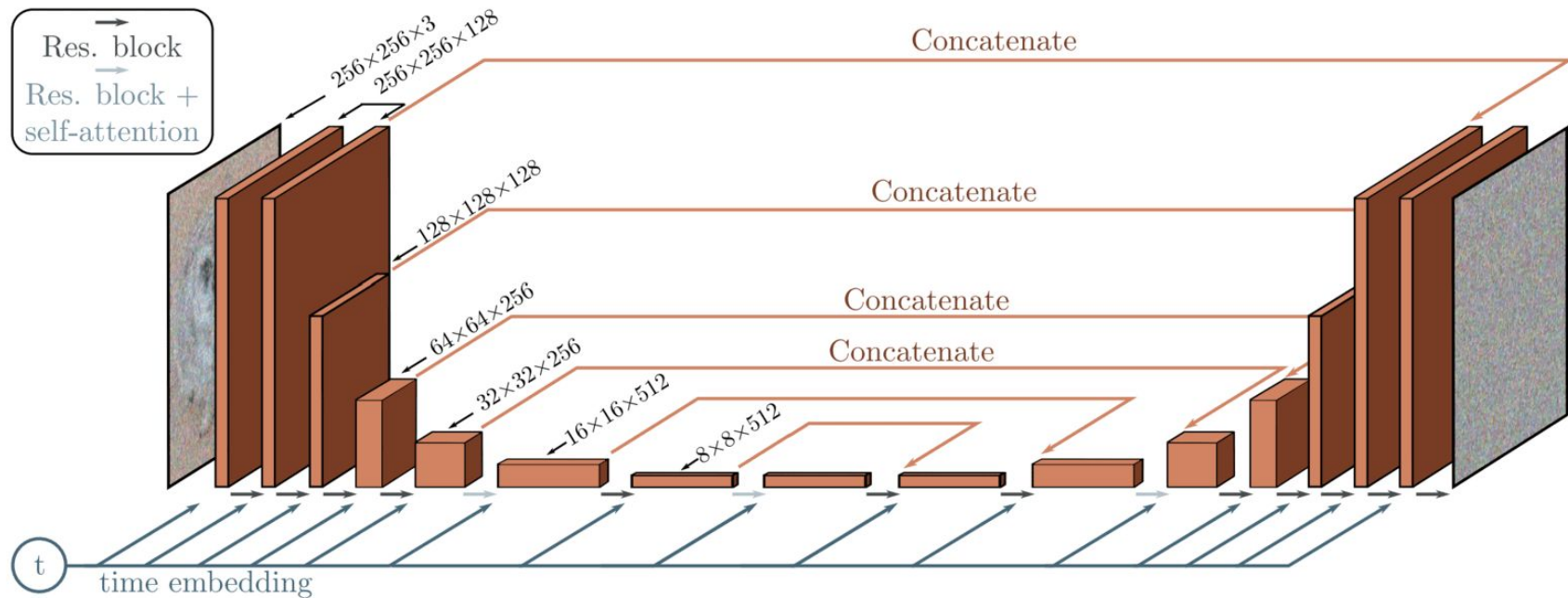
Training Objective: Mean-squared error (MSE) between the predicted noise and the actual added noise added by a random scheduler.

Sampling: Then you run the learned reverse diffusion steps repeatedly, denoising gradually until you get a realistic image



Denoising Diffusion Probabilistic Models (DDPM)

In DDPMs, the authors use a UNet-shaped deep neural network:



Denoising Diffusion Probabilistic Models (DDPM)

TorTolSe's DDPM is conditioned on:

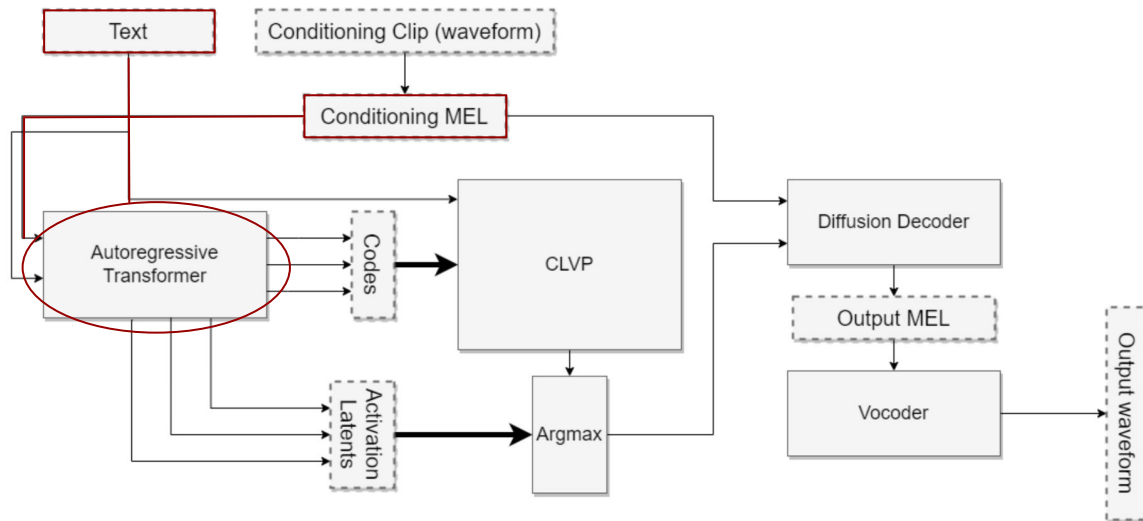
- Diffusion step
- Generated speech codes from the GPT-2 model
- Reference mel-spectrogram from the target voice

The **TorTolSe** trick:

Since the autoregressive transformer and the DDPM are trained separately, fine-tuning the latter on the transformer's output latents, couples the 2 components, significantly boosting the quality and increasing the efficiency.

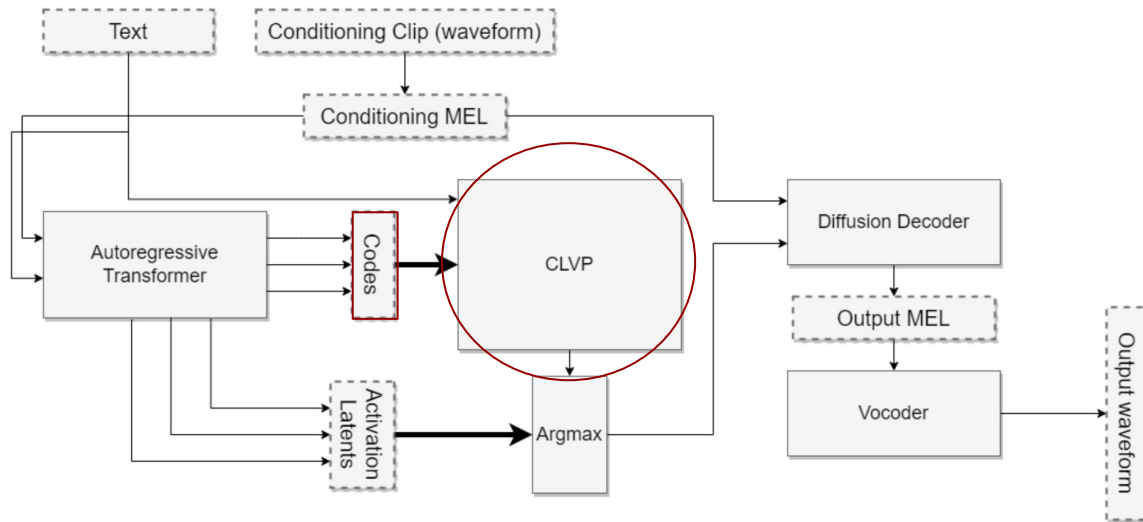
TorToiSe - How it works?

The autoregressive decoder predicts a probability distribution for speech tokens, conditioned on text and reference speaker mel-spectrograms.



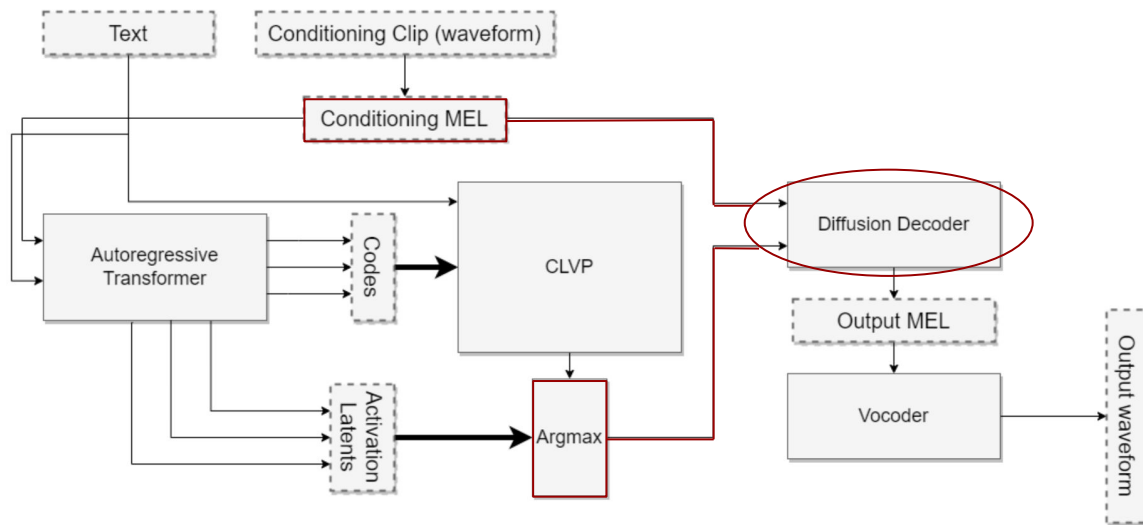
TorToiSe - How it works?

The CLVP module serves as a scoring model to re-rank the speech codes from the AR model



TorToiSe - How it works?

The diffusion-based decoder predicts the speech mel-spectrogram from the predicted speech codes and the conditional mel-spectrogram.



TorToiSe - How it works?

Finally, the Univnet Vocoder converts mel-spectrogram into waveforms.

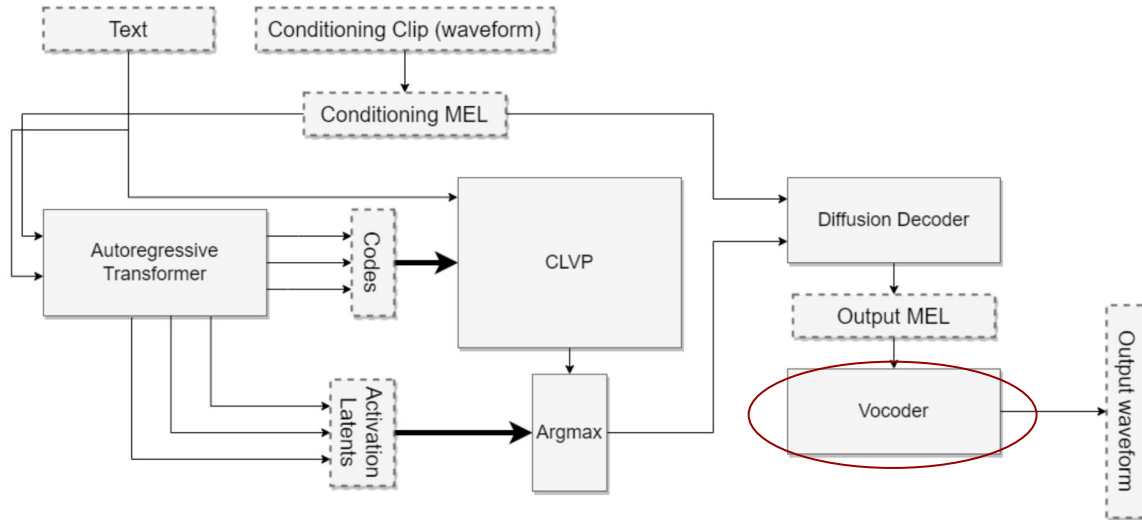


Table of contents

Introduction

TorToiSe

Scaling TTS

Universal audio generation

Scaling TTS - Overview

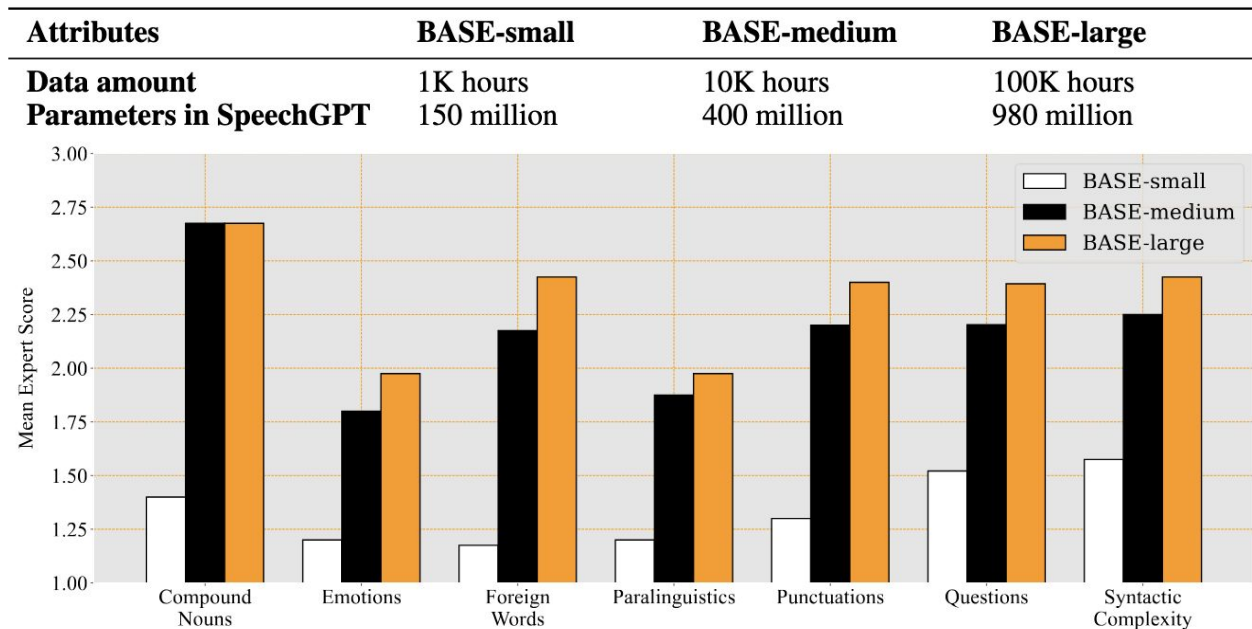
	Current Systems	VALL-E
Intermediate representation	mel spectrogram	audio codec code
Objective function	continuous signal regression	language model
Training data	≤ 600 hours	60K hours
In-context learning	X	✓

[Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers](#), Microsoft, Jan 2023

“In-context learning” is a prompt-based approach for zero-shot TTS, which does not require additional structure engineering, pre-designed acoustic features, and fine-tuning.

Control emotions, acoustic environment and identity with just a prompt.

Scaling TTS - Effects



Scaling improves “emergent behaviours”

Scaling TTS - Complexity

Is it just a matter of scaling?

Scaling data and models was crucial to improve the traditional TTS systems:

- Larger datasets cover more speakers, prosodies, accents and styles
- Larger models achieves more realistic results and benefit from more compute

However, computing was already present, and while large transcribed speech datasets are smaller than text/image ones, a similar domain like Automatic Speech Recognition leverages already hundreds of thousands of hours.

Scaling TTS - Complexity

Can we just scale parameters and data of Tacotron2 or FastSpeech2?

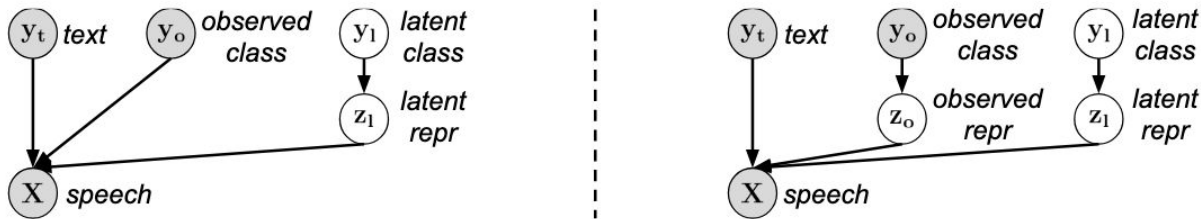
No! They don't model the right distribution. The MSE loss applied on mel-spectrograms leads to underfitting and average not realistic outputs.

- TTS is a one-to-many problem, while ASR is mostly many-to-one in clean speech scenarios. Also humans are very sensitive to unnatural sounds.
- Minimizing MSE is mathematically equivalent to maximizing the likelihood of data generated by a Gaussian process, however speech is far more complex.

Scaling TTS - Complexity

Previous works attempted to solve these issues:

- Labelling training data with multiple attributes to condition the model and/or learn disentangled representations → expensive and hard to scale
- Learning latent representation similar to GST or using a VAE to learn the unobserved variation $p(\text{speech} \mid \text{latent}, \text{text})$ → difficult to control and train



Scaling TTS - Model families

Currently, there are 2 main model families that overcome underfitting:

Examples of models

- **Token-based**, discretize the audio and use cross-entropy loss to learn complex distributions

GPT-2, Llama, Parti, VALL-E, BASE
TTS, SoundStream, AudioGen,
MusicGen, ...

- **Gradient-based**, like diffusion or flow models, that learn to map from simple to complex distributions

DALL-E, Stable Diffusion
NaturalSpeech2, VoiceBox,
AudioBox, AudioLDM, ...

Table of contents

Introduction

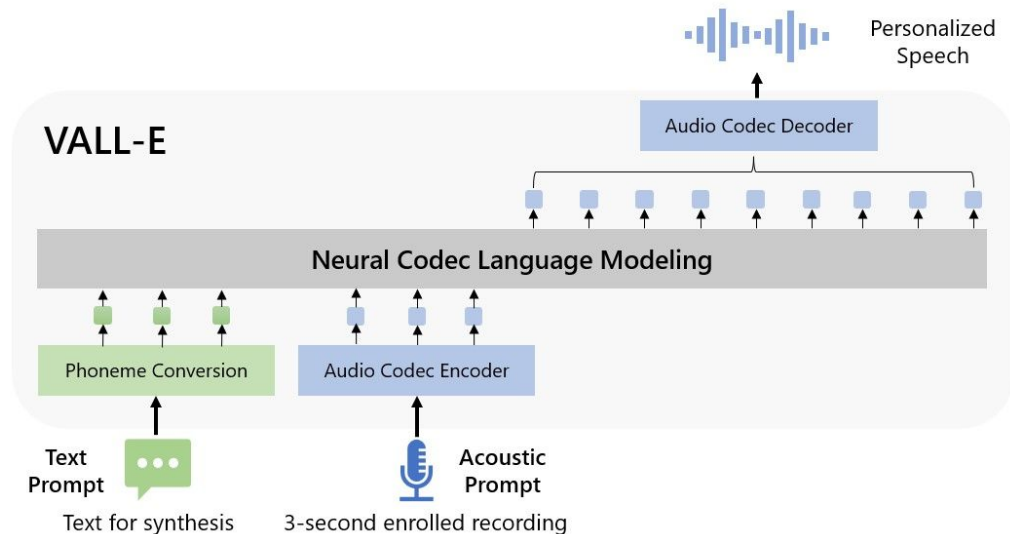
TorToiSe

Scaling TTS

Universal audio generation

Universal audio generation - Current design

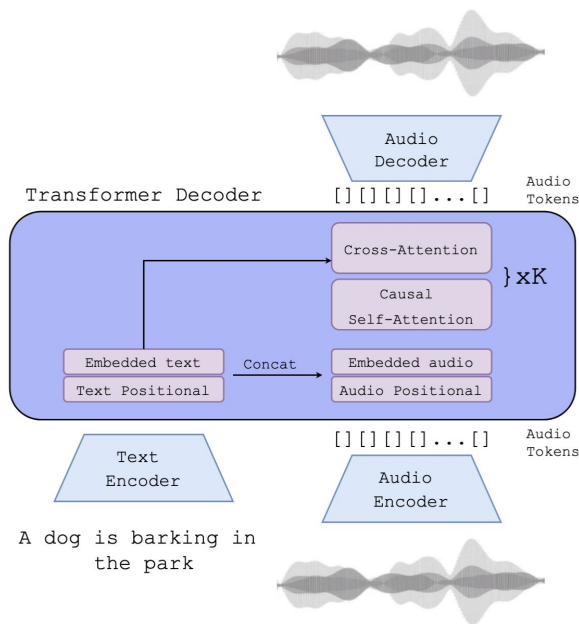
VALL-E simplifies TorToiSe's design leveraging Encodec to both encode the reference speech and decode the generated speech.



[Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers](#), Microsoft, 2023

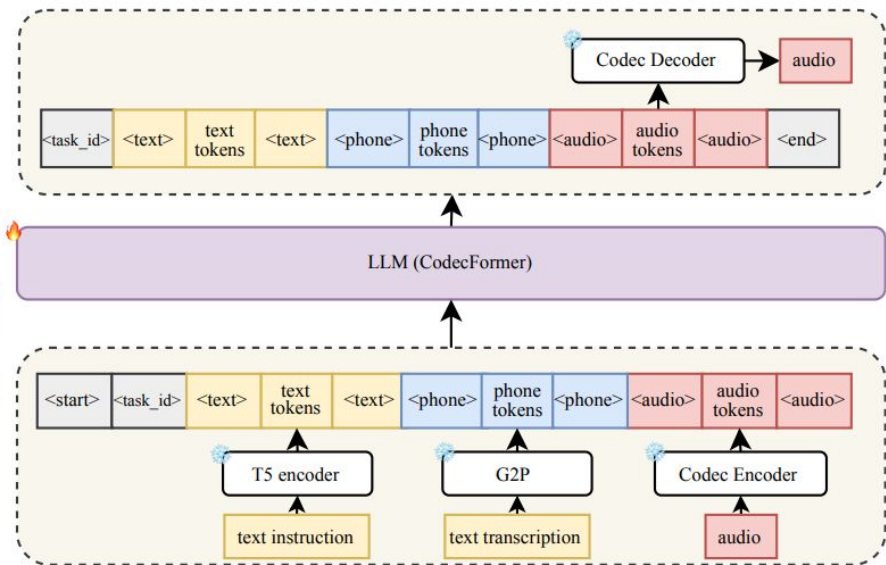
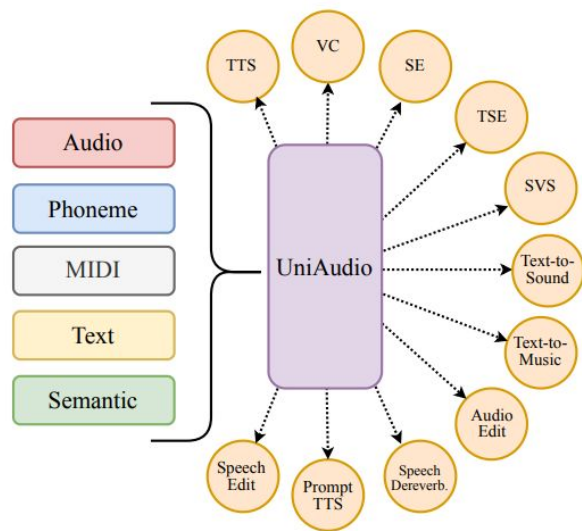
Universal audio generation - Audio and music

The same paradigm can be applied for other domains like music and audio generation.



Universal audio generation - Multi-task

As the models generalize and the representation unifies we start to consider the possibility to leverage LLMs to solve multiple tasks.



Universal audio generation - Challenges

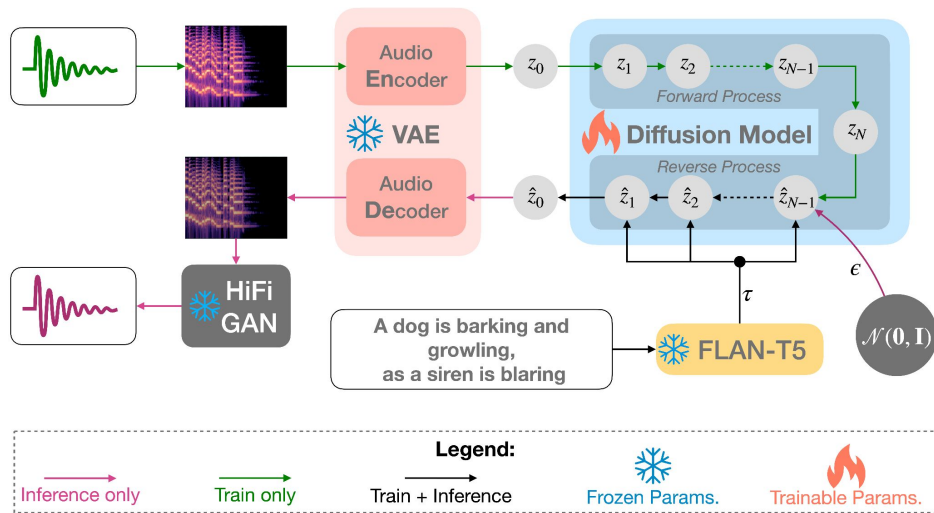
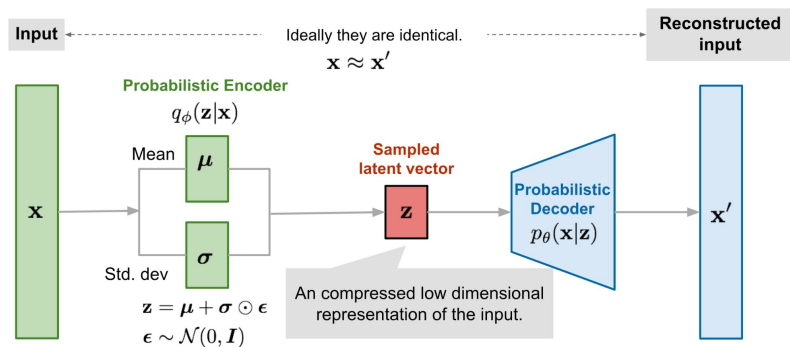
In practice, modeling speech and audio as discrete codes is not as simple as it might seem, and there are still several challenges:

- 2 stage training, speech reconstruction vs representation
- Semantic meaning of RVQ codes and inconsistency issues
- Modeling multiple-codebooks (quality/performance trade-offs)

Thank you

Universal audio generation - Extra

There are several models that uses simpler continuous representation without introducing quantization reconstruction bottlenecks and ad-hoc design.

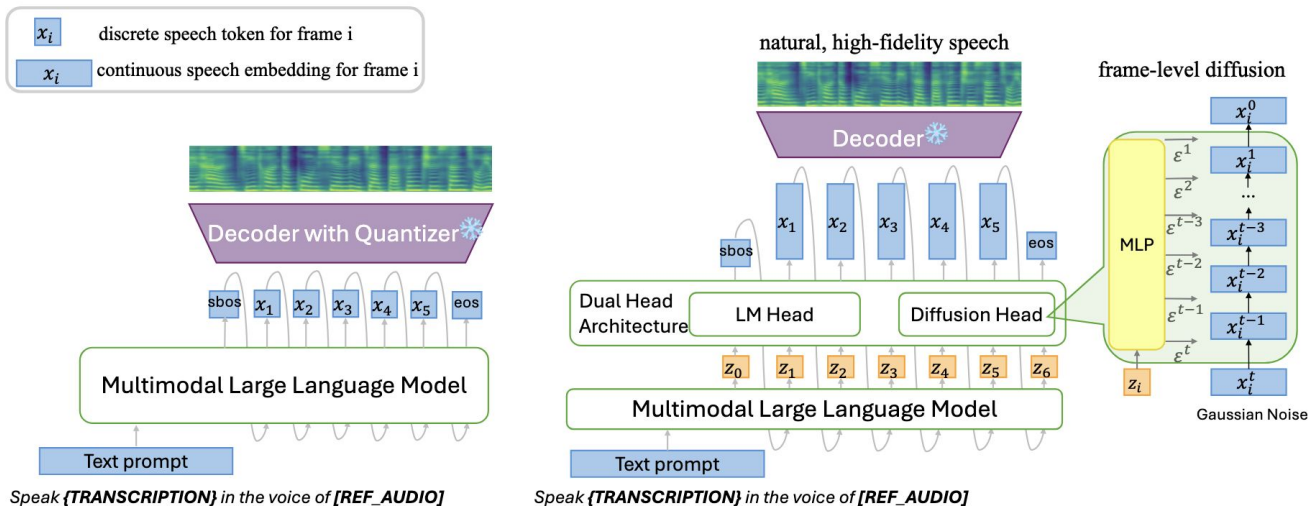


VAE architecture, from blog.bayeslabs.co

[Text-to-Audio Generation using Instruction-Tuned LLM and Latent Diffusion Model](#), (2023)

Universal audio generation - Extra

Recently, autoregressive LLMs have been trained to predict directly continuous representations using small diffusion heads, marrying the two approaches.



(a) Discrete Speech Generation in MLLM

(b) Continuous Diffusion Speech Generation in MLLM